

OPIM 201 Operations Management

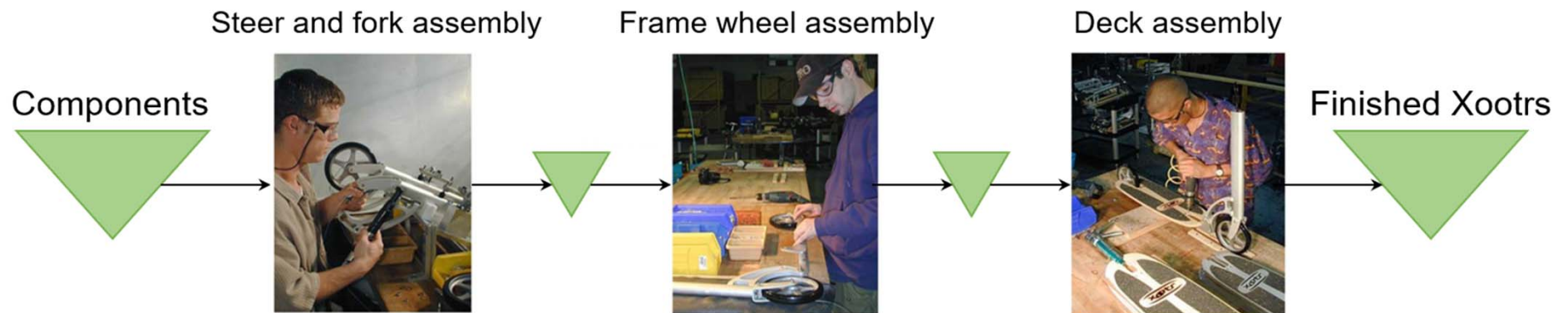
INVENTORY MANAGEMENT: EOQ MODELS (C&T Ch. 5)



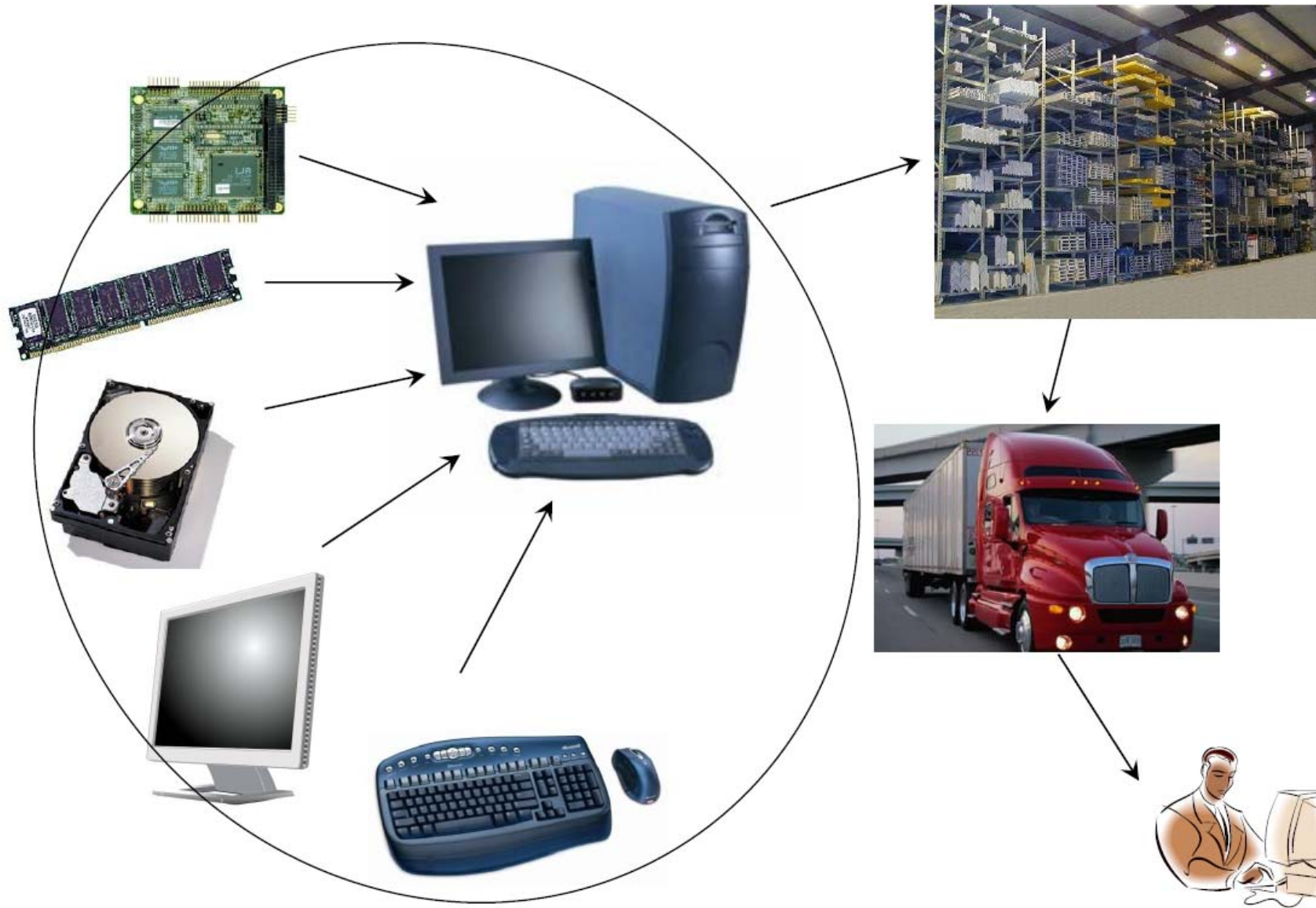
Inventory

- Inventory is any stored resource used to satisfy need.
- Examples: Cash in a bank, blood supply in a hospital, water in a reservoir, and finished products in a warehouse.
- Inventory classification:
 - Raw materials: has been purchased, but not processed.
 - Work-in-process: has been processed, but not completed.
 - Finished products: completed products, waiting shipment.

Xootr process flow diagram

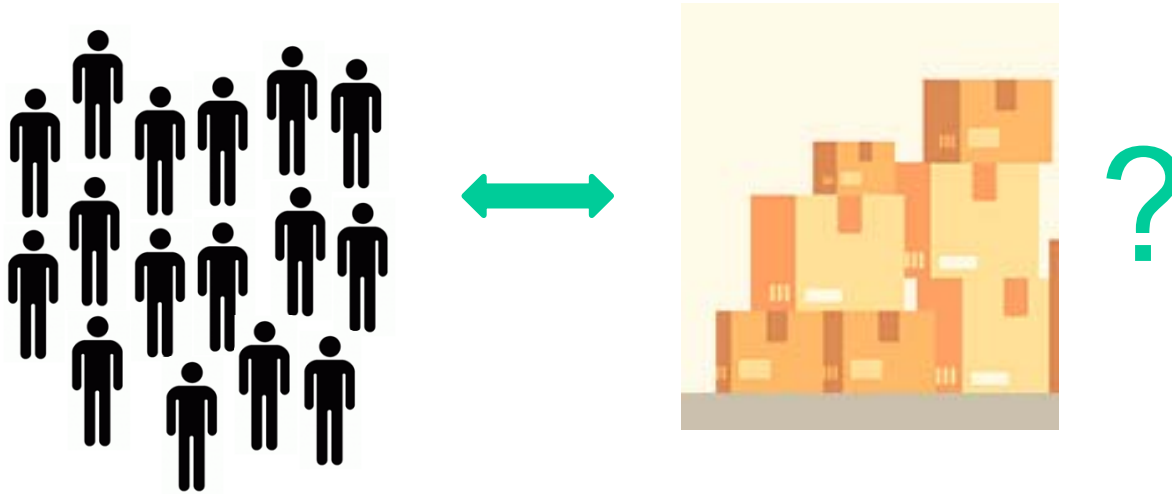


Inventory Example



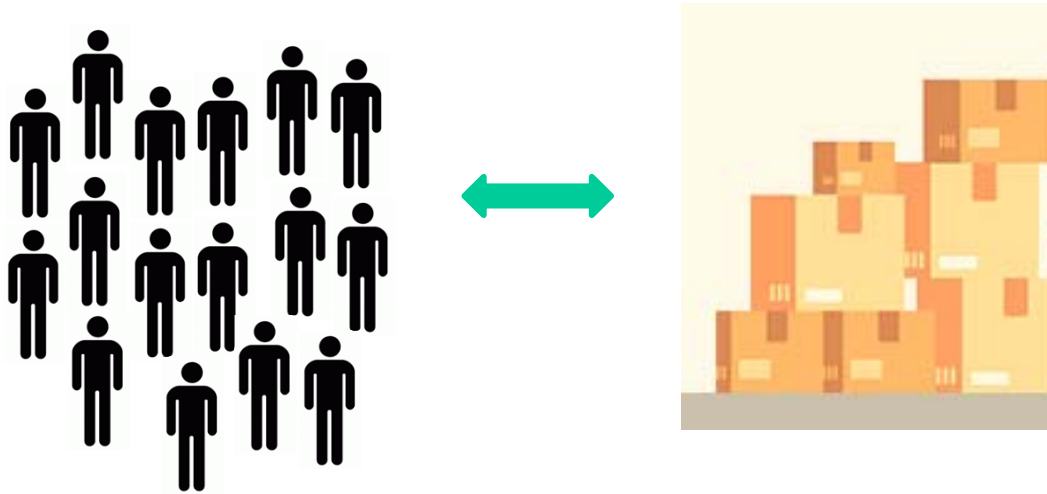
Another Classification of Inventory

- The average daily demand is 100 units. How many units would you order?



Another Classification of Inventory

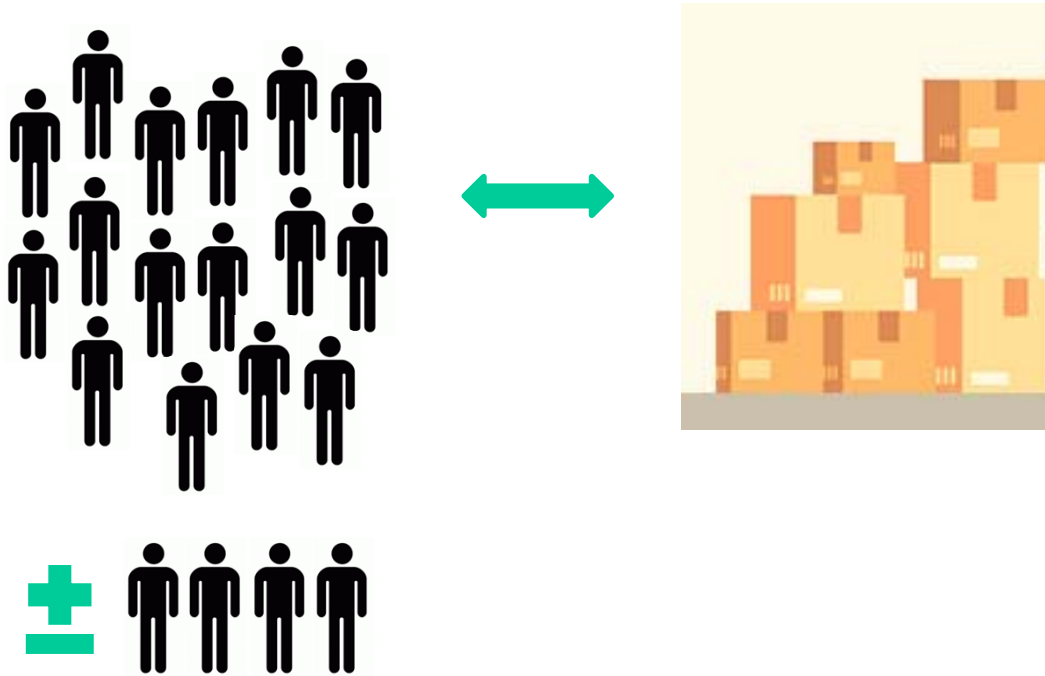
- The average daily demand is 100 units. How many units would you order?



- Cycle stock: meet average demand during an ordering cycle.

Another Classification of Inventory

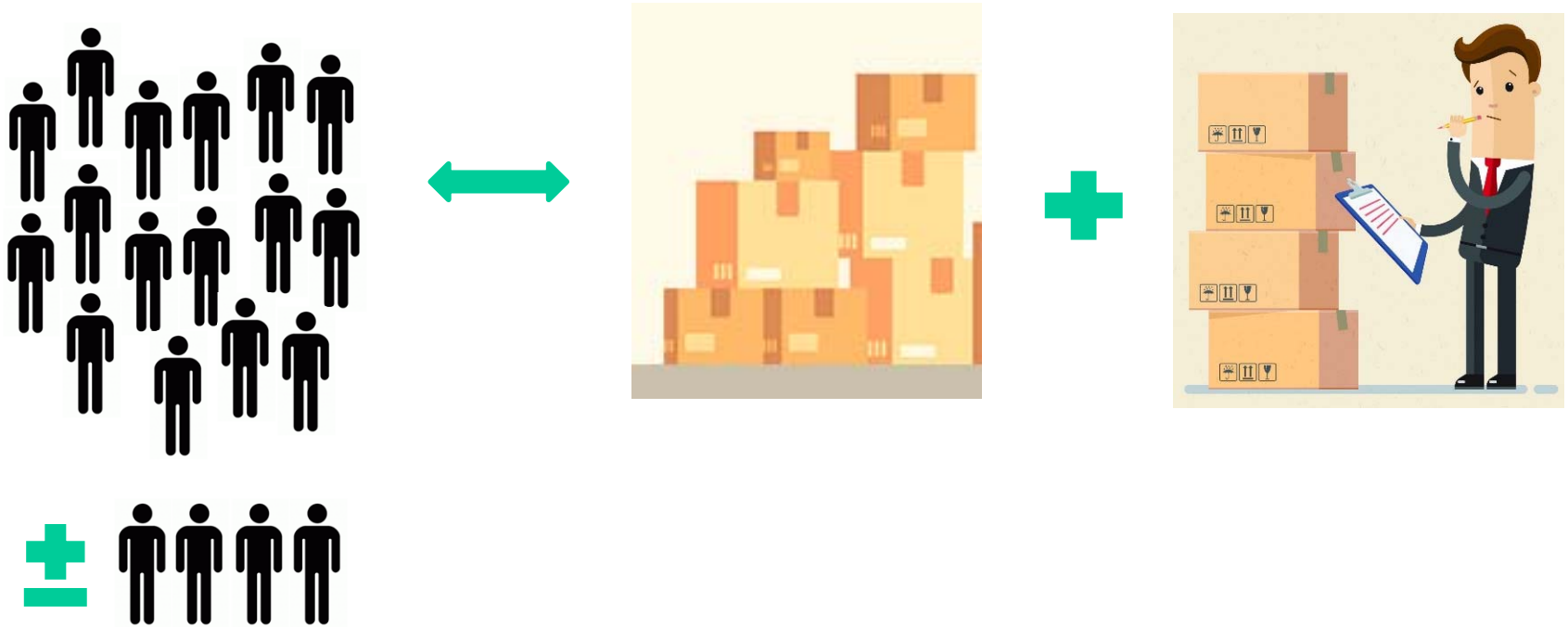
- The average daily demand is 100 units. How many units would you order?



- Cycle stock: meet average demand during an ordering cycle.

Another Classification of Inventory

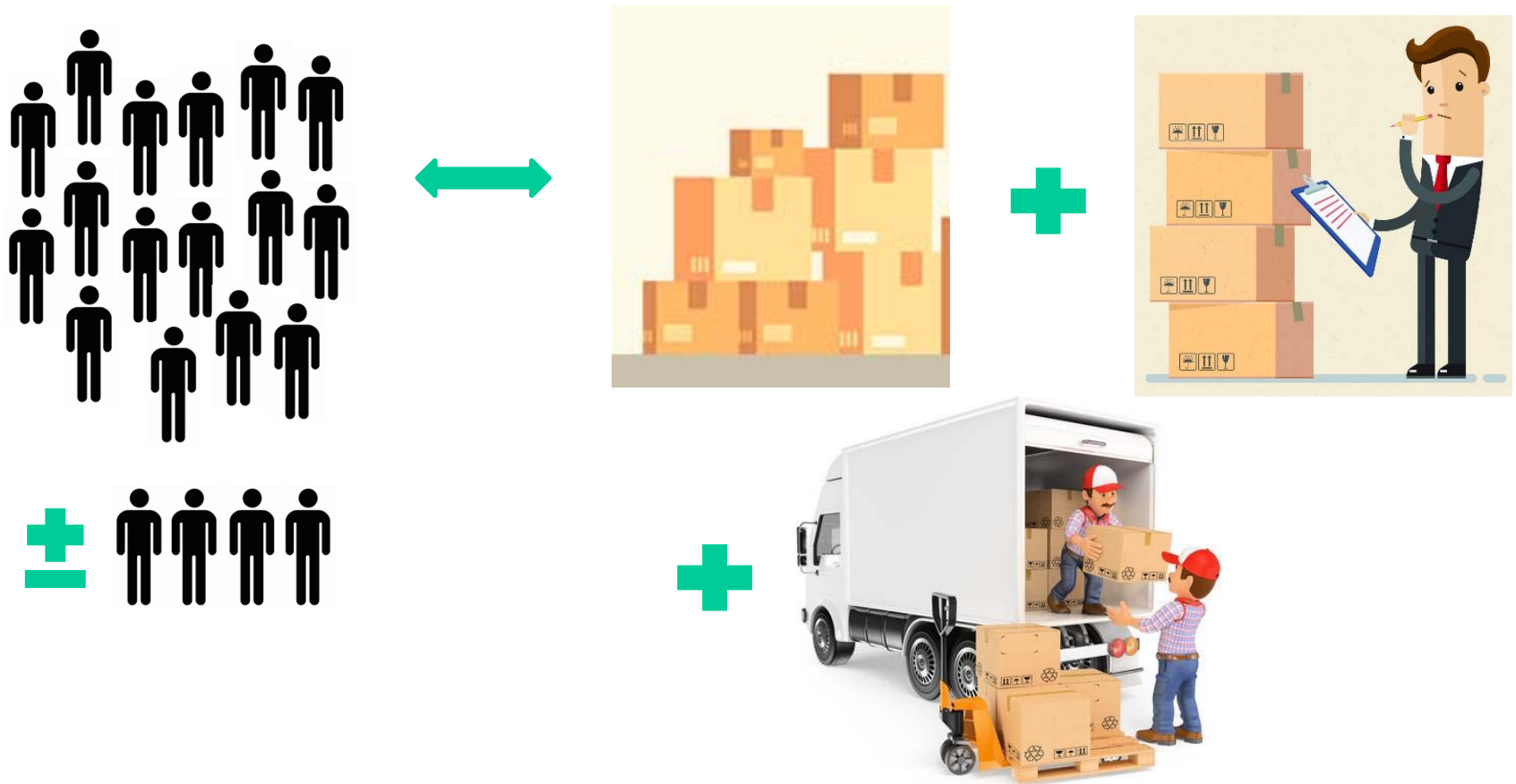
- The average daily demand is 100 units. How many units would you order?



- Cycle stock: meet average demand during an ordering cycle.
- Safety stock: inventory carried in addition to the expected demand.

Another Classification of Inventory

- The average daily demand is 100 units. How many units would you order?

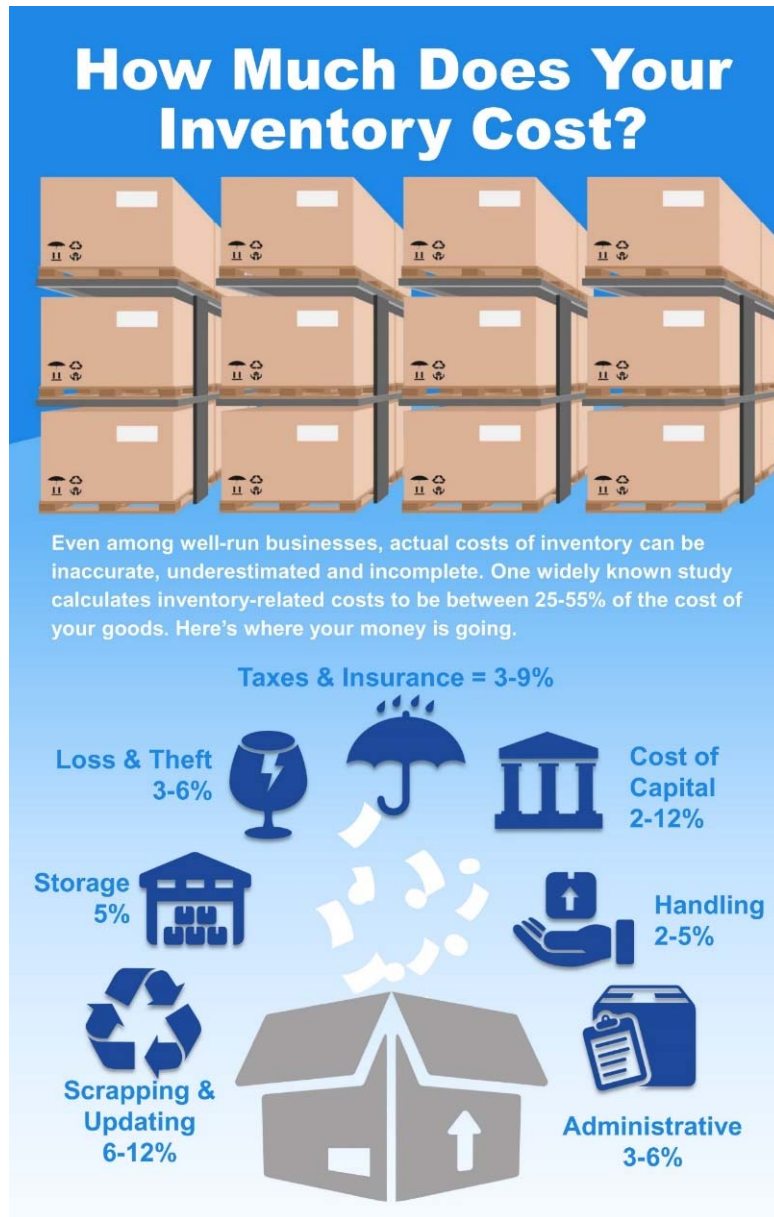


- Cycle stock: meet average demand during an ordering cycle.
- Safety stock: inventory carried in addition to the expected demand.
- Pipeline inventory: orders that have been placed but not yet received.

Amazon.com

- Amazon.com started as a “virtual’ retailer – no inventory, no warehouses, no overhead; just computers taking orders to be filled by others
- Amazon now stocks millions of items of inventory in 69 warehouses around the world
- Growth has forced Amazon.com to become a world leader in warehousing and inventory management
- Amazon.com obtains competitive advantage through inventory management

Why is Inventory Management Important?



- Inventory: one of the most expensive assets of many companies.
- Represents as much as 25-55% of total invested capital.
- Example:
 - Currently: If a firm carries an inventory of \$20 million, it costs more than \$6 million/year (~30%)
 - Through better inventory management: If the inventory reduces to \$10 million, then save over \$3 million.
- The objective of inventory management is to find a balance between inventory investment and customer service.

Inventory Policy

- An Inventory Policy determines when stock should be replenished, and how large orders should be.
 - Order Timing: when to order
 - Order Quantity: how much to order

- Models covered in this course:
 - Economic Order Quantity (EOQ)
 - Newsvendor (next week)

- More advanced models in the following courses:
 - Supply Chain Management
 - High Performance Warehousing and Fulfilment

Components of Inventory Cost

- Holding costs
 - e.g., storage facilities, handling, insurance, tax, pilferage, obsolescence, cost of capital, etc.
 - Include any cost components that are proportional to the amount of time a flow unit spends in the system.
 - Typically proportional to the value of the inventory, i.e., can be expressed as a percentage of the acquisition cost of the units.
- Backordering costs or Shortage costs (for delaying or canceling orders)
 - e.g., lateness penalty, lost profit, goodwill loss
- Production or Procurement costs (in general acquisition costs)
 - Fixed costs
 - Ordering or setup (e.g., clerical, equipment setups, transportation)
 - Independent of the amount you produce or procure
 - Incur only if you produce or place an order (unlike sunk costs)
 - Variable purchase or production cost
 - Proportional to the amount you acquire

Example 1: Ordering Handle Caps for the Xootr

Novacruz, a kick scooter producer, sources handle caps for its scooters from a Taiwan supplier for \$2 per unit. Currently, the inventory is depleted at the rate of the assembly operations, that is, at a flow rate of 500 units per week. A shipping company charges Novacruz a fixed price of \$1000 for shipping any amount up to 15000 units. The annual unit holding cost is 40% of the unit purchase price. Assuming there are no other costs, how many handle caps should they order each time they order from their supplier?

- \$2 = cost to Nova Cruz to purchase each handle cap (a pair) from its supplier in Taiwan.
- \$1000 = fixed shipping fee (including customs fee) for each shipment, independent of the amount ordered.
- 500 = demand for handle caps per week.
- 40% = Nova Cruz's annual inventory holding cost



Economic Order Quantity (EOQ) Model

Assumptions:

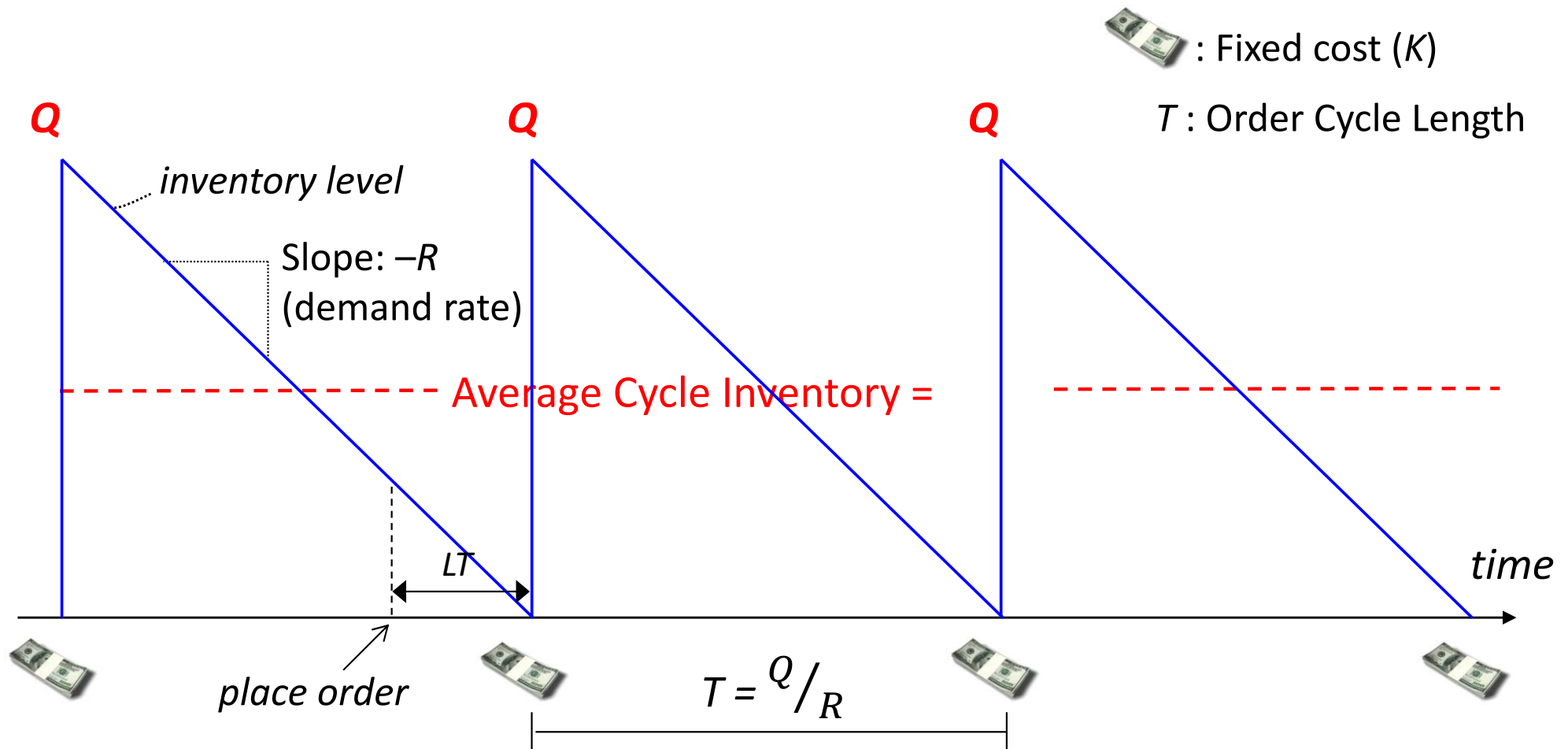
- Constant & known demand rate / flow rate (e.g., R units per year)
- Inventory never spoils, degrades or is lost; everything is eventually sold
- Orders are delivered with a reliable lead time
 - Lead time (LT): time between when an order is placed & when it is received
- Unit purchase price is independent of the quantity purchased

Costs:

- Unit variable acquisition (procurement or production) cost: p
- Unit holding cost per unit of time: h (typically expressed as % of p)
- Fixed costs per order: K

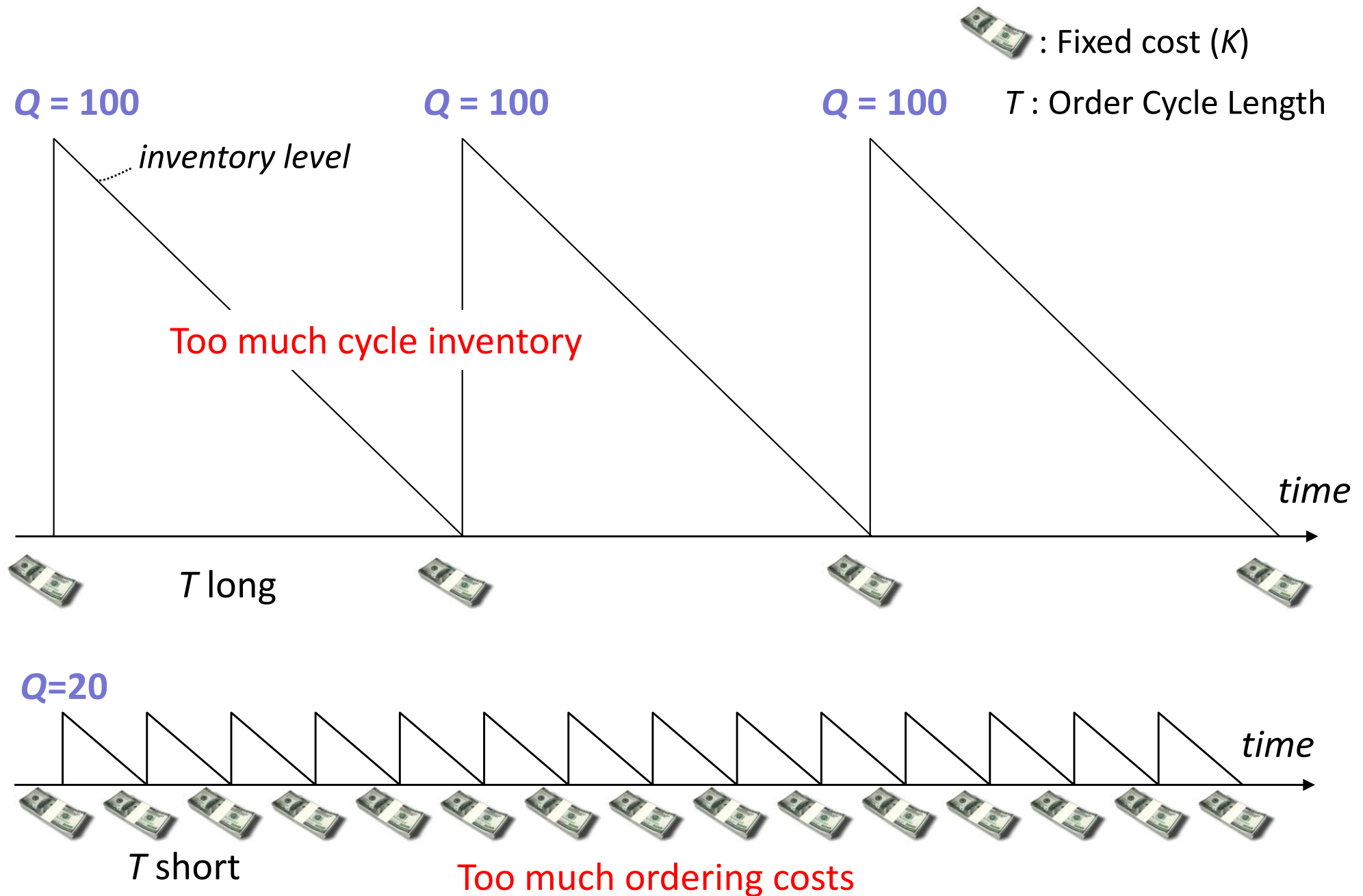
Decision: Order Quantity Q
(What about Order Timing?)

Trade-Off: Fixed Costs vs. Cycle Inventory Costs



- As $Q \uparrow$, inventory costs $\uparrow$ but fixed cost $\downarrow$.
- The optimal Q (or T) balances inventory holding costs and ordering costs.
- For positive lead time ($L > 0$), order L periods before inventory level becomes zero.

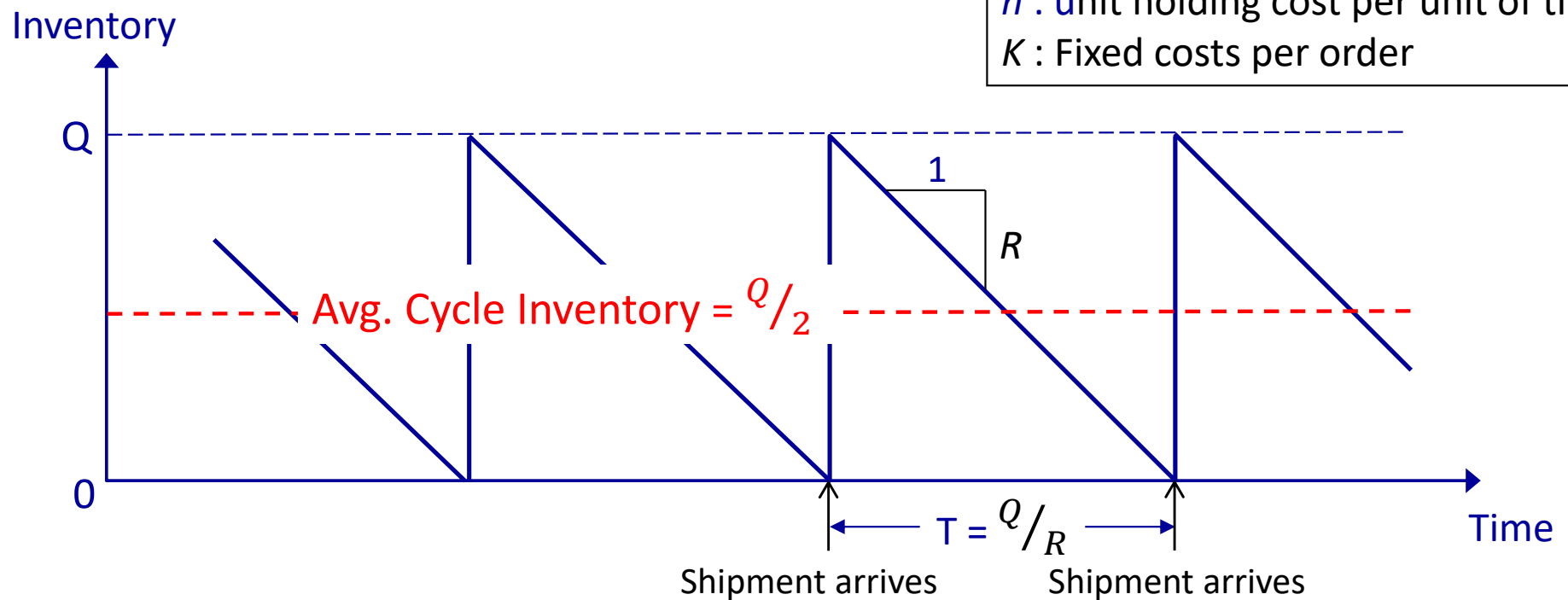
Trade-Off: Fixed Costs vs. Cycle Inventory Costs



Trade-Off: Fixed Costs vs. Cycle Inventory Costs

- Very often, employees focus on one cost (perhaps due to limitation of their role/position or the incentive system), say fixed costs and end up ordering too much and too infrequently (to reduce ordering cost).
- They should balance the two costs by assessing how their order quantity impact these two costs. How?

The Economic Order Quantity (EOQ), Q^*



Goal: Determine the economic order quantity Q^* that minimizes inventory holding cost and fixed cost per unit time, $C(Q)$

$C(Q)$ = Inventory holding costs per unit time + Fixed costs per unit time

$$= h \times Q/2 + K \times \left(\frac{R}{Q} \right) \leftarrow \text{Number of orders per unit time}$$

Understanding the Computation of Fixed Costs

- Fixed costs per unit time = $K \times \frac{R}{Q}$ ← Number of orders per unit time
 OR $K / (\frac{Q}{R}) = K/T$

Example

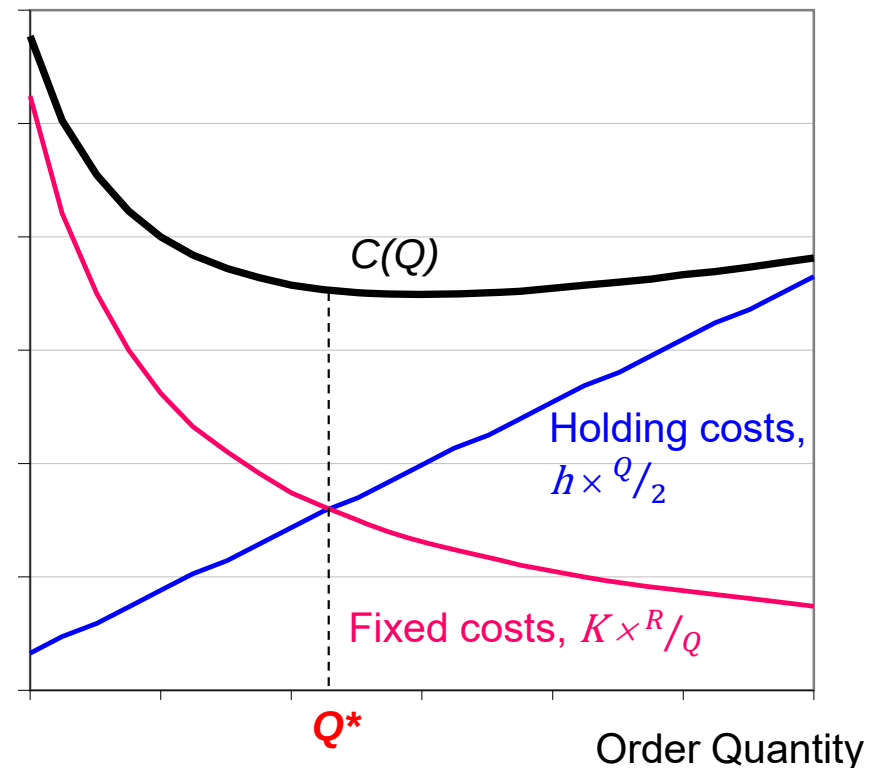
- $R = 100$ units/month (= 1200 units/yr.)
- $Q = 200$
- $T = Q/R = 200/100 = 2$ month (or $200/1200 = 1/6$ yr.)
- Number of orders per year = $1/T = R/Q = 1200/200 = 6$ times per year (or $100/200 = 1/2$ times per month). Hence fixed costs per year is $K \times R/Q = 6K$
- Alternatively, since a fixed cost K is assessed once per order cycle, it is charged to the duration of the order cycle, T , which is 2 months or $1/6$ year. Hence the fixed cost per unit time (during each order cycle) is $K/T = K/(\frac{Q}{R})$. In particular, the fixed cost per year is $K/(1/6) = 6K$.

The Economic Order Quantity (EOQ), Q^*

- Objective: Choose Q to minimize $C(Q) = h \times Q/2 + K \times R/Q$
 - Purchasing cost ($R \times p$) is independent of the order decision
 - The total cost, $TC(Q) = R \times p + h \times Q/2 + K \times R/Q$
- Solution: Order the Economic Order Quantity, Q^* :

$$Q^* = \sqrt{\frac{2 \times R \times K}{h}}$$

$$\begin{aligned} C(Q^*) &= \frac{Q^*}{2} h + \frac{R}{Q^*} K \\ &= \sqrt{2RK h} \end{aligned}$$



Check Your Understanding – Example 1

- Purchase costs:
 - \$2 per unit x 500 per week = \$1000 per week
 - Q cannot influence our weekly purchase cost!
- h = Inventory holding cost per unit time:
 - 40% annual holding cost, so ...
 - $0.4 \times \$2 = \0.80 = cost to hold a unit for one year...
 - $h = \$0.80 / 52 = \0.01538 = cost to hold a unit for one week.
- K = Ordering cost:
 - $K = \$1000$; this is the fixed shipping fee per order that is independent of the amount ordered as long as it does not exceed 15000.

$$Q^* = \sqrt{\frac{2 \times R \times K}{h}} = \sqrt{\frac{2 \times 500 \times 1000}{0.01538}} = 8062$$

$$C(Q^*) = \sqrt{2RK h} = \sqrt{2(500)(1000)(0.01538)} = \$124.03 \text{ per week}$$

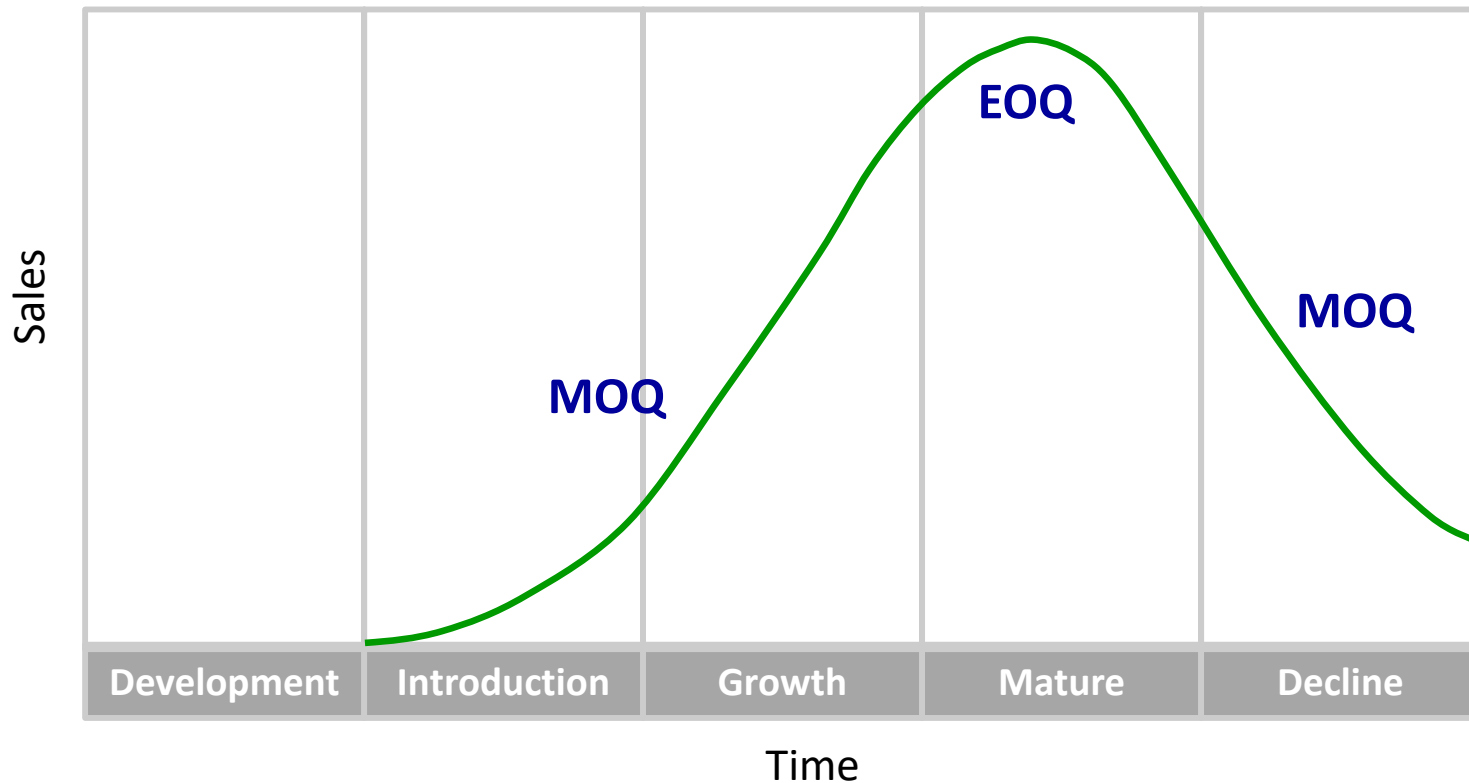
$$T = Q^*/R = 8062/500 \approx 16 \text{ weeks or 4 months}$$

Discuss: Challenges in Applying EOQ in Practice

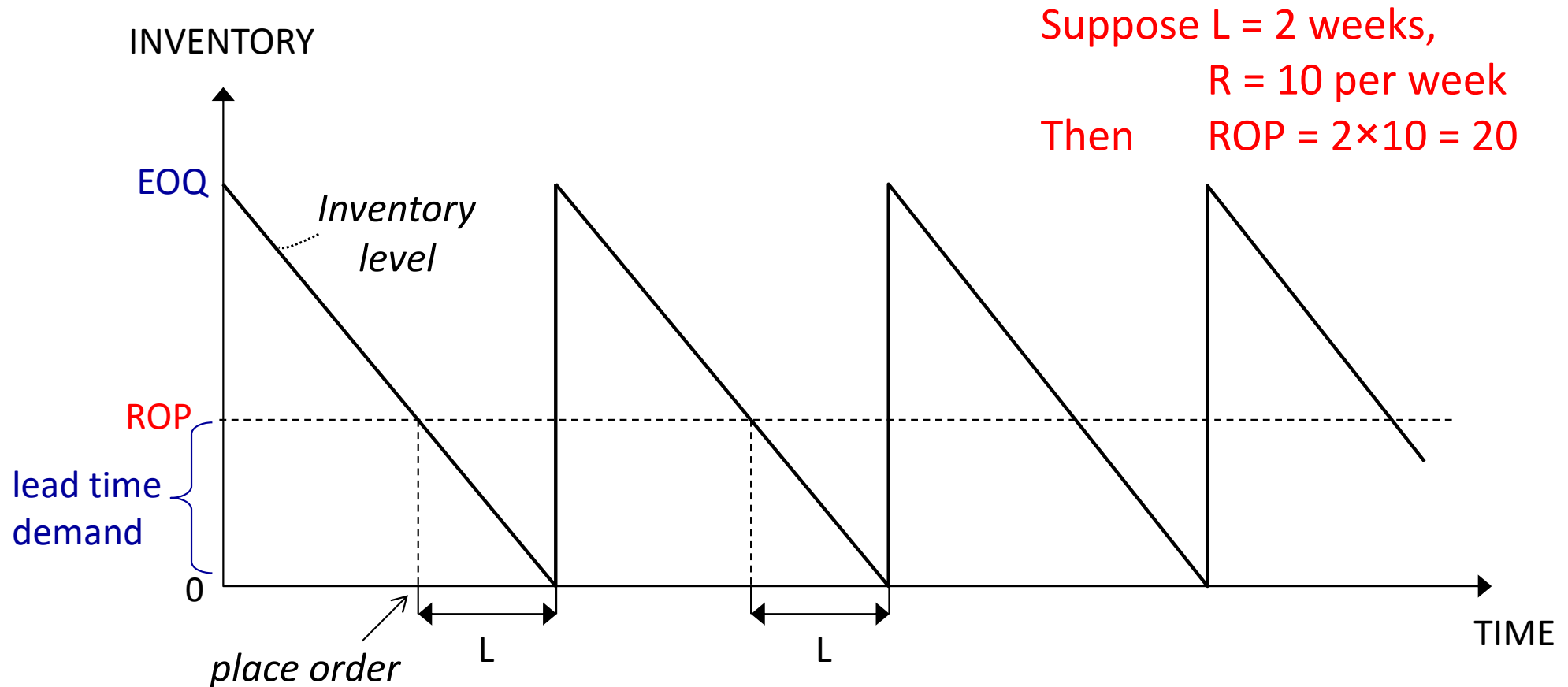
- Suppliers impose Minimum Order Quantity (MOQ)
 - Reason: fixed cost incurred by suppliers or to facilitate storage and transportation (e.g., inventory measured in pallet rather than in unit)
 - Lead to high inventory cost at customers
 - Infrequent order from customer can also cause problem in capacity planning for upstream suppliers – constant continuous consumer demand translates into lumpy infrequent demand for suppliers
 - Could be resolved via collaboration and technology
- Demand is uncertain
- Difficult to estimate cost parameters (h and K)

Addressing Demand Uncertainty

- EOQ are usually used in the mature stage of product life cycle due to lower risk of obsolescence
- In the case when demand uncertainty is significant, EOQ are used together with ROP system (where safety stock inventory may be incorporated in the ROP)



Impact of Lead Time: The Reorder Point (ROP) System



- Place an order when inventory position reaches a Re-Order Point, ROP

$$ROP = \text{demand during lead-time} = R \times L$$

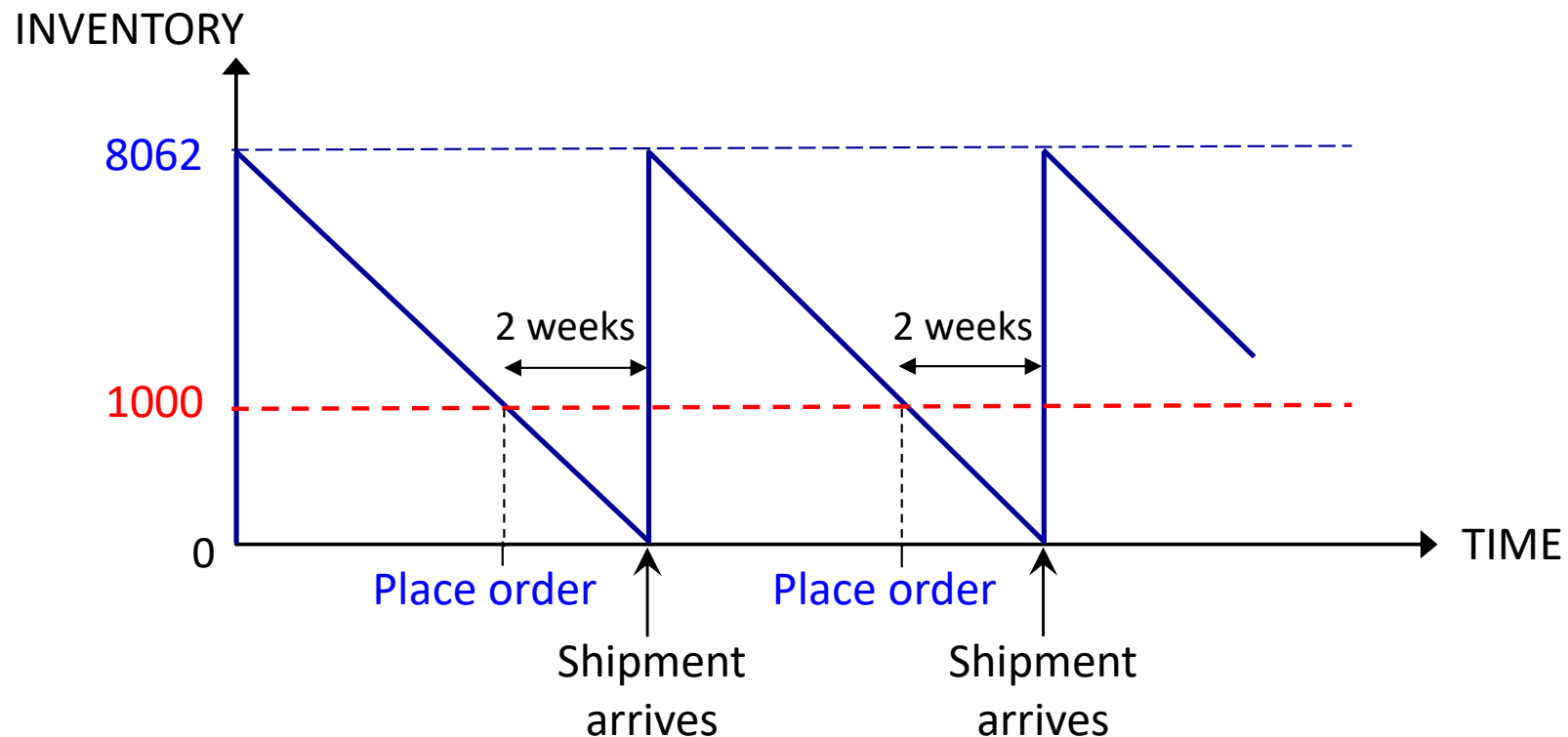
where R = demand rate; L = supply lead-time

Example 1 – Revisit

- Suppose the original conditions in Example 1 continue to hold. Moreover, it takes two weeks for an order to arrive from Taiwan. What is the ordering policy, that is, order-size and reorder point?

$$\text{Order EOQ} = \sqrt{\frac{2 \times R \times K}{h}} = \sqrt{\frac{2 \times 500 \times 1000}{0.01538}} = 8062$$

$$\text{Reorder Point} = 500 \text{ caps per week} \times 2 \text{ week} = 1000$$



Check Your Understanding: Example 2

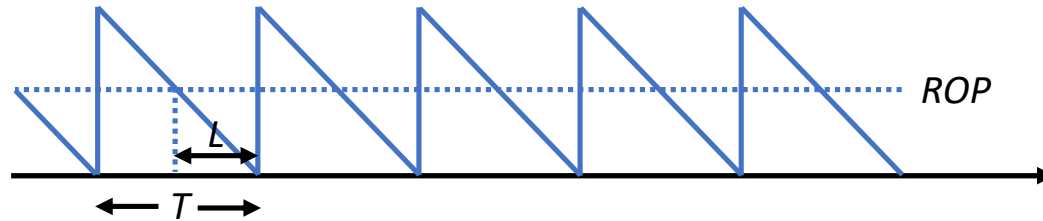
A local distributor for Handi-Man Company expects to sell approximately 30,000 two-wheel carts of a certain size and design next year. The distributor's cost of placing an order with the manufacturing facility is \$300 and the manufacturer's price for each cart is \$25. The annual cost for carrying inventory is 30 percent of the item purchase price. The distributor operates 250 days per year. The manufacturer's lead time for replenishing this product is five working days. Answer the following questions.

- (a) What is the economic order quantity (EOQ)?

- (b) What is the reorder point (ROP) in units for the distributor based on the above information?

Impact of Lead Time: Inventory Position

- So far, we have assumed in the EOQ model that the delivery lead time, L is zero or is less than the order cycle length, T (determined by the EOQ).



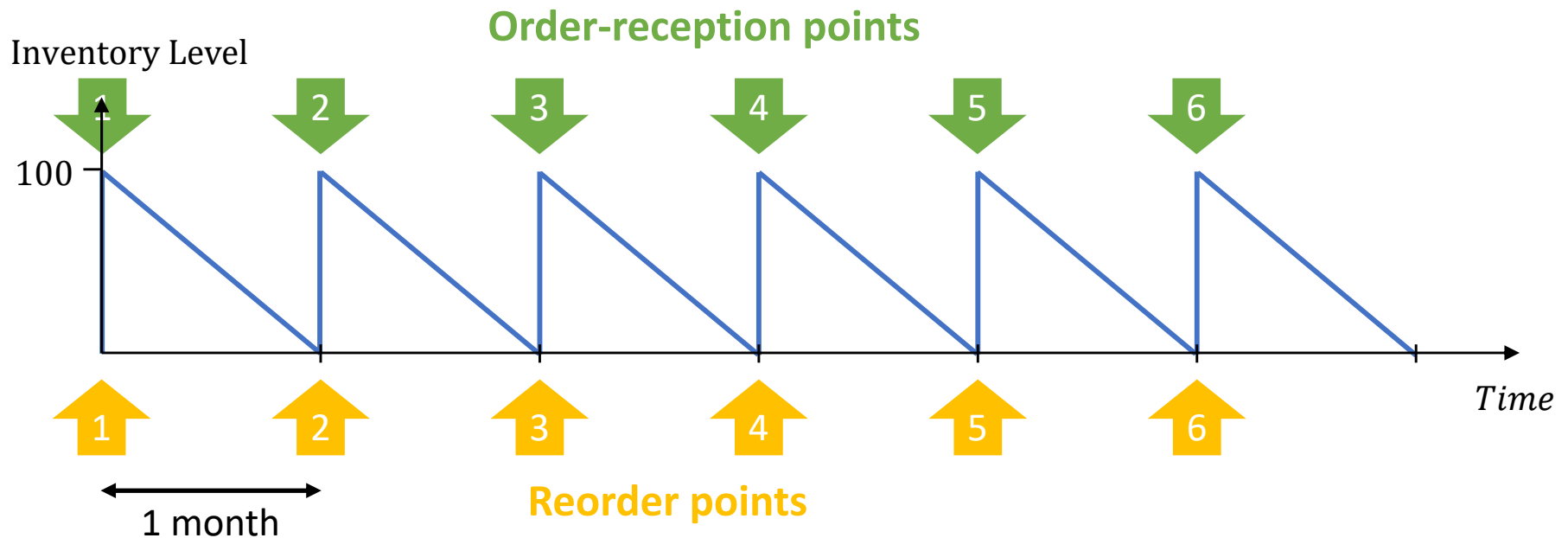
- What is the ordering policy if L is greater than T ?
- Define

Inventory Position (IP) = On-hand Inventory + Pipeline inventory

- Recall that safety stock is zero in the EOQ model. Given:
 - Q^* = optimal order quantity suggested by the EOQ model.
 - R = demand rate.
 - L = delivery lead time.
- Policy:
 - Order Q^* units when inventory position $IP = R \times L$ (ROP).

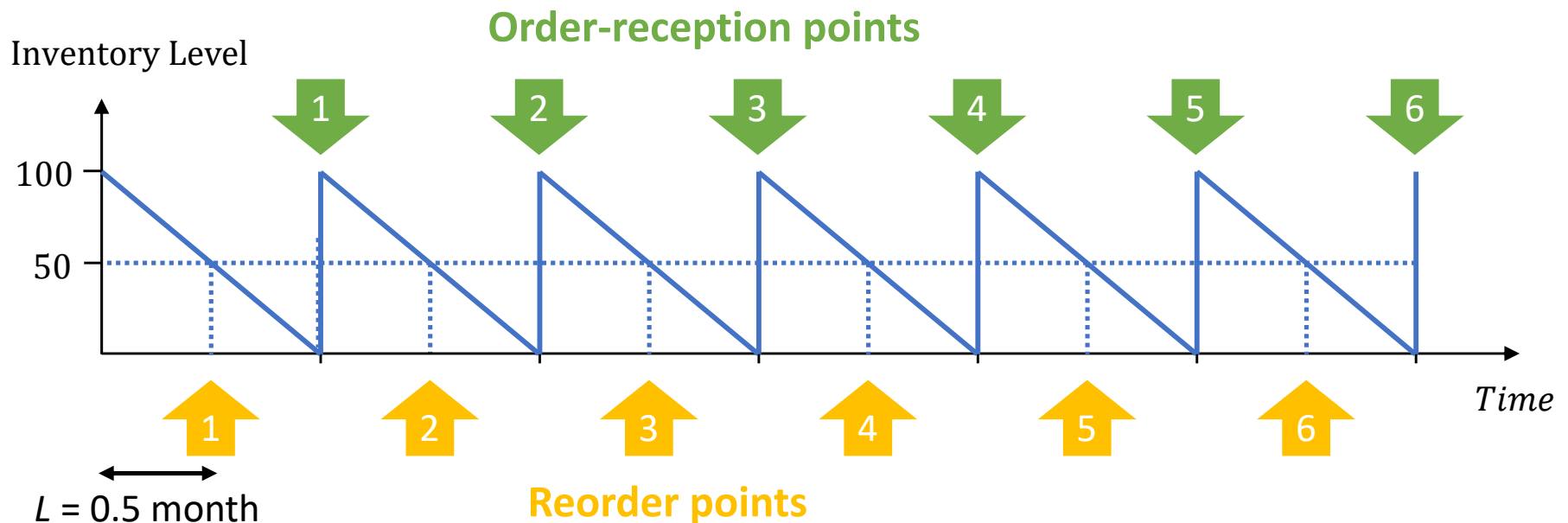
Impact of Lead Time: $L = 0$

- Suppose the demand rate is 1200 units per year and the optimal order quantity is 100 units (order cycle length = one month).
- Suppose lead time $L = 0$:
 - According to ordering policy: order Q^* when $IP = 1200 \times 0 = 0$ units.
 - We reorder only when we exhaust all the inventory on hand.
 - Reordering points are same as the points when the orders are received.



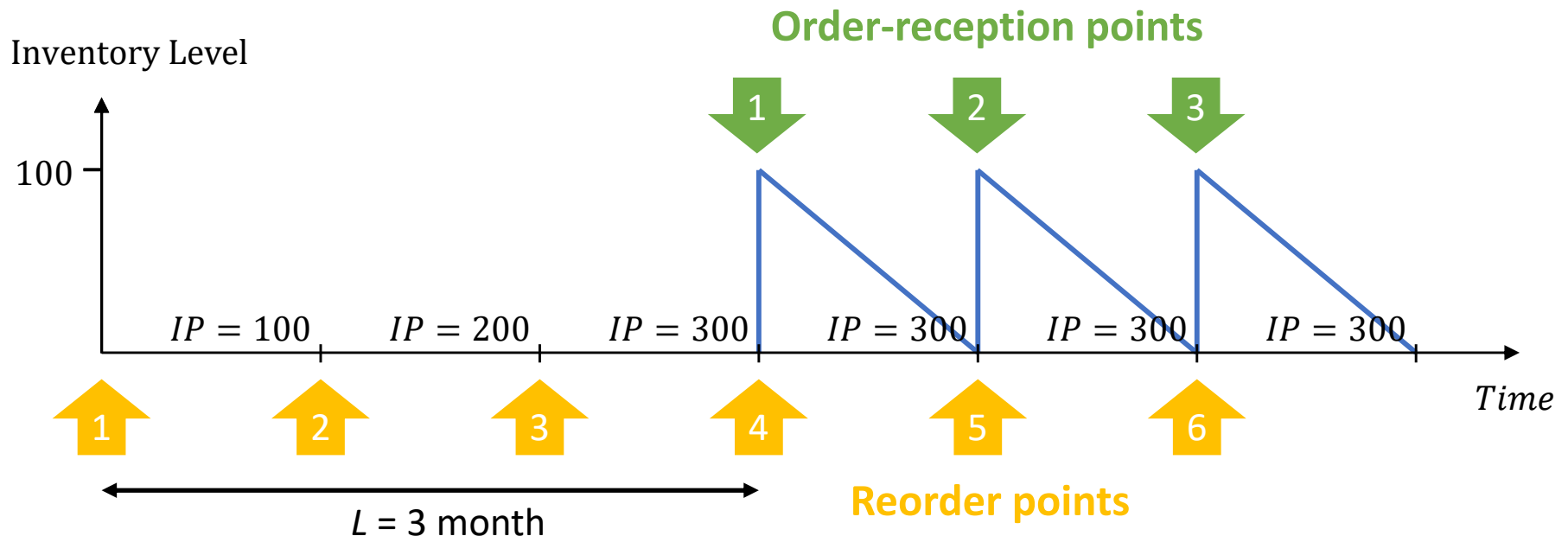
Impact of Lead Time: $L \leq T$

- Suppose the demand rate is 1200 units per year and the optimal order quantity is 100 units (order cycle length = one month).
- Suppose lead time $L =$ half month:
 - According to ordering policy: order Q^* when $IP = (1200 \text{ units/year}) \times 0.5 \text{ month} = (1200 \text{ units/year}) \times (0.5/12) \text{ year} = 50 \text{ units}$.
 - We reorder when the on-hand inventory falls to 50 units.
 - Reordering points and the order-reception points are separated by half month duration.



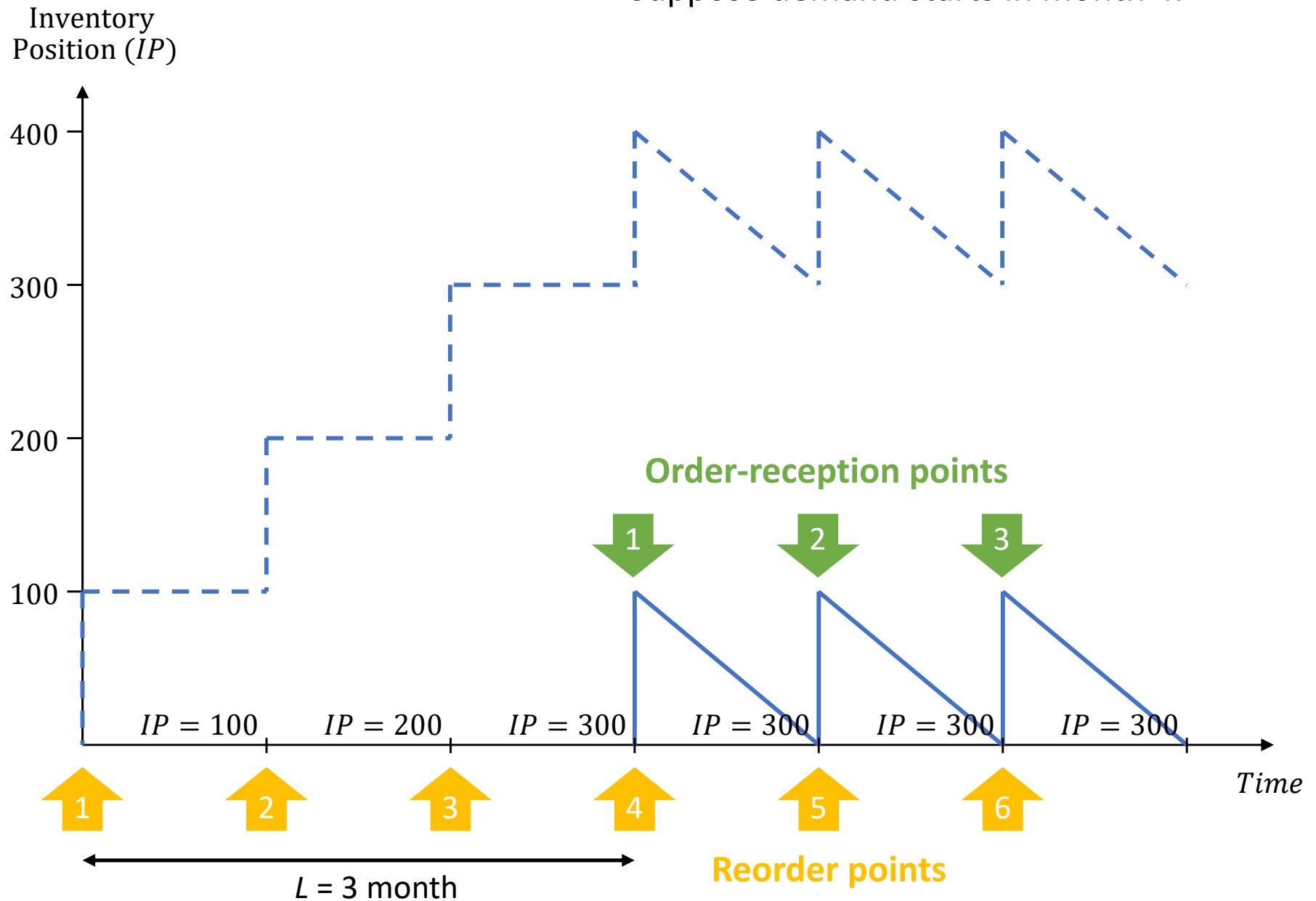
Impact of Lead Time: $L > T$

- Suppose the demand rate is 1200 units per year and the optimal order quantity is 100 units (order cycle length = one month).
- Suppose lead time $L =$ three months:
 - According to ordering policy: order Q^* when $IP = (1200 \text{ units/year}) \times 3 \text{ months} = (1200 \text{ units/year}) \times (3/12) \text{ year} = 300 \text{ units}$.
 - We reorder when the inventory position falls to 300 units.
 - Reordering points and the order-reception points are separated by three months duration.



Impact of Lead Time: $L > T$

- Suppose demand starts in month 4.



Robustness of EOQ

- EOQ model requires accurate estimates of parameters such as K & h . How do inaccurate estimates of these parameters (leading to incorrect computation of Q^*) affect the system performance?
- *EOQ* model is very robust:

For any Q ,

$$\frac{C(Q)}{C(Q^*)} = \frac{1}{2} \left(\frac{Q}{Q^*} + \frac{Q^*}{Q} \right)$$

- Example: Suppose $Q = 2Q^*$.

$$\frac{C(Q)}{C(Q^*)} = \frac{1}{2} \left(2 + \frac{1}{2} \right) = 1 \frac{1}{4}$$

Hence, if Q is twice as much as Q^* , $C(Q)$ only increases by 25%.

Sensitivity Analysis

$$Q^* = \sqrt{\frac{2 \times R \times K}{h}}$$

$$C(Q^*) = \sqrt{2RK h}$$

- Q^* :
 - If K decreases (by n times), then Q^* decreases (by $\sqrt{n}$ times).
 - If h decreases (by n times), then Q^* increases (by $\sqrt{n}$ times).

- $C(Q^*)$:
 - If K decreases (by n times), then $C(Q^*)$ decreases (by $\sqrt{n}$ times).
 - If h decreases (by n times), then $C(Q^*)$ decreases (by $\sqrt{n}$ times).

Check Your Understanding

- 1) Natural is a firm that produces different flavors of jam, each with a constant demand. Natural manages its inventory for each flavor using an EOQ model. It has signed a three-year contract to rent a warehouse at \$1000 per month. The warehouse is used to store the jam produced. How does this contribute to the cost structure under the EOQ model?
 - a) Holding Cost
 - b) Fixed Cost
 - c) Shortage Cost
 - d) Variable Production Cost
 - e) None of the above

- 2) Natural is highly automated and its machinery is depreciating at a rate of \$100 per day. How does the depreciating cost contribute to the cost structure under the EOQ model?
 - a) Holding Cost
 - b) Fixed Cost
 - c) Shortage Cost
 - d) Variable Production Cost
 - e) None of the above

Key Lesson

- Do not confuse cost accounting (or activity-based costing – allocating costs for accounting purposes) with identifying relevant cost parameters for making operational decision
 - E.g., some costs such as salaries or wages are actually sunk costs and should not be charged as fixed costs or holding costs.

Summary

- Inventory management:
 - Helps to coordinate flows across supply chain
 - Keeps costs low, and keeps customers satisfied
- Ordering too much at one time:
 - Reduces administrative and fixed costs of orders
 - Increases inventory holding costs
- Ordering too little at one time:
 - Reduces inventory holding costs, but increases fixed and administrative costs
- EOQ model
 - Balances order and holding costs
 - Order with moderate frequency
- Time between successive orders: depends on optimal quantity